Variable-Projection Curvature for Efficient Newtonized Refinement in Wideband Direct Position Determination

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ABSTRACT

Unknown wideband waveforms couple source positions to many frequency coefficients in direct position determination. Block Newtonized orthogonal matching pursuit reduces the search dimension, but position updates based on fixed coefficients can make slow progress even when waveforms are re-estimated during acceptance. We develop a variable-projection curvature update for this local refinement. Eliminating the current source coefficients removes a positive-semidefinite coupling term from the position Hessian, and a local convergence analysis explains the resulting small-step reduction. Frequency-wise accumulation requires only small matrices. In 240 paired two-source observations, replacing fixed-coefficient curvature under the same waveform-eliminated acceptance objective reduces mean complete runtime by 83.77%, with identical success/failure decisions at a 20 m tolerance. A further 480-observation test distinguishes this local efficiency gain from the configuration sensitivity of the complete sequential locator. The method improves the computational efficiency of block refinement without changing residual-based source discovery.

1. Introduction

Direct position determination (DPD) fits synchronized receiver observations jointly to estimate emitter positions, avoiding intermediate delay association [1, 5]. For overlapping unknown waveforms, positions and frequency coefficients must be estimated together. Sequential source discovery reduces the spatial search, but its cost can be dominated by repeated local refinement.

Newtonized orthogonal matching pursuit (NOMP) alternates discovery, Newton refinement and cyclic feedback [4]. Multi-frequency block NOMP extends this construction to two-dimensional positions and unknown source coefficients at each frequency [6]. Its fixed-coefficient position derivatives provide the closest refinement reference. QNOMP jointly updates delay and angle parameters using quasi-Newton optimization and gain re-estimation in pilot-based channel estimation [7]. Here we retain sequential block updates and study the exact local curvature for unknown waveforms.

Variable projection eliminates linear coefficients before nonlinear optimization [2]. We specialize its curvature to wideband DPD, show how position–waveform coupling can slow fixed-coefficient updates, and accumulate the correction with small frequency-wise matrices. The experimental comparison keeps waveform-eliminated acceptance and numerical controls common to isolate the curvature change. This

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distinction matters because original NOMP already re-estimates gains in its acceptance test [4]; re-estimation alone is not the contribution.

2. Model and local objective

Let $\boldsymbol{r}_l \in \mathbb{R}^2$ and $\boldsymbol{p}_k \in \Omega \subset \mathbb{R}^2$ denote station l and source k. For L synchronized single-antenna stations and known source count K, the retained DFT observations satisfy

$$\boldsymbol{x}_f = \sum_{k=1}^{K} \boldsymbol{a}_f(\boldsymbol{p}_k) s_{k,f} + \boldsymbol{n}_f, \qquad [\boldsymbol{a}_f(\boldsymbol{p})]_l = \frac{\exp[-\mathrm{j}2\pi f d_l(\boldsymbol{p})/c_0]}{d_l(\boldsymbol{p})}, \tag{1}$$

where $d_l(\boldsymbol{p}) = \|\boldsymbol{p} - \boldsymbol{r}_l\|$, c_0 is propagation speed, $f \in \mathcal{F}$ is signed baseband frequency, and $F = |\mathcal{F}|$. The unknown $s_{k,f} \in \mathbb{C}$ includes transmitted power. The known relative amplitude follows free-space loss; its common wavelength factor is absorbed into $s_{k,f}$ [3]. We use a static, calibrated, direct-path baseband-delay model, without RF carrier phase or random link gains. The common-window approximation neglects station-dependent record edges [1]. Independent complex Gaussian sample noise of variance σ_t^2 has retained-bin variance σ_t^2/N for a $1/N$-normalized DFT of length N. Numerical evaluation floors distances at 1 m; the following smooth derivatives apply outside that protection region.

Write $E = \sum_f \|\boldsymbol{x}_f\|^2 > 0$. During refinement of source j, hold the other fitted contributions fixed and form $\boldsymbol{y}_f = \boldsymbol{x}_f - \sum_{k\neq j} \boldsymbol{a}_f(\boldsymbol{p}_k)\widehat{s}_{k,f}$. For trial position $\boldsymbol{z} \in \mathbb{R}^2$,

$$\mathcal{L}(\boldsymbol{z}, \boldsymbol{q}) = E^{-1} \sum_f \|\boldsymbol{y}_f - \boldsymbol{a}_f(\boldsymbol{z}) q_f\|^2, \qquad \widehat{q}_f(\boldsymbol{z}) = \frac{\boldsymbol{a}_f^H \boldsymbol{y}_f}{\boldsymbol{a}_f^H \boldsymbol{a}_f}, \tag{2}$$

$$\phi(\boldsymbol{z}) = \min_{\boldsymbol{q}} \mathcal{L}(\boldsymbol{z}, \boldsymbol{q}) = E^{-1} \sum_f \left(\|\boldsymbol{y}_f\|^2 - \frac{|\boldsymbol{a}_f^H \boldsymbol{y}_f|^2}{\boldsymbol{a}_f^H \boldsymbol{a}_f} \right). \tag{3}$$

Here $\boldsymbol{x}_f, \boldsymbol{y}_f, \boldsymbol{n}_f \in \mathbb{C}^L$ and $\boldsymbol{q} \in \mathbb{C}^F$. Superscripts H and T denote conjugate and ordinary transpose; $\mathrm{j}^2 = -1$ and I_2 is the identity. All current-source coefficients are eliminated; the other source contributions remain fixed within this local call.

3. Waveform-eliminated curvature

At the current position, set $q_f = \widehat{q}_f$, $\boldsymbol{e}_f = \boldsymbol{y}_f - \boldsymbol{a}_f q_f$, and define $\boldsymbol{a}_{f,i} = \partial \boldsymbol{a}_f / \partial z_i$ and $\boldsymbol{a}_{f,ij} = \partial^2 \boldsymbol{a}_f / \partial z_i \partial z_j$. Differentiating with fixed coefficients gives

$$g_i = -\frac{2}{E} \operatorname{Re} \sum_f \boldsymbol{e}_f^H \boldsymbol{a}_{f,i} q_f, \tag{4}$$

$$[H_{zz}]_{ij} = \frac{2}{E} \operatorname{Re} \sum_f \left(|q_f|^2 \boldsymbol{a}_{f,i}^H \boldsymbol{a}_{f,j} - \boldsymbol{e}_f^H \boldsymbol{a}_{f,ij} q_f \right). \tag{5}$$

Coefficient optimality makes $\boldsymbol{g}$ also the gradient of ϕ, but H_{zz} is not its Hessian. To obtain the reduced Hessian, collect coefficient real and imaginary parts in $\boldsymbol{u} \in \mathbb{R}^{2F}$. Differentiating $\nabla_u \mathcal{L}(\boldsymbol{z}, \widehat{\boldsymbol{u}}(\boldsymbol{z})) = 0$ gives

$$\frac{\partial \widehat{\boldsymbol{u}}}{\partial \boldsymbol{z}} = -H_{uu}^{-1} H_{uz}, \qquad H_{\mathrm{VP}} = H_{zz} - H_{zu} H_{uu}^{-1} H_{uz}. \tag{6}$$

This standard elimination identity [2] requires a smooth response and a locally unique coefficient minimizer. All blocks are evaluated at the current coefficient optimum. Here H_{uu} is block diagonal with 2×2 blocks $2\|\boldsymbol{a}_f\|^2 I_2 / E$, so it is positive definite. Consequently, $H_{zz} - H_{\mathrm{VP}} \succeq 0$: fixed-coefficient curvature includes response changes that waveform adjustment can compensate.

For explicit evaluation, let $\boldsymbol{D}_f = [\boldsymbol{a}_{f,1}, \boldsymbol{a}_{f,2}]$, $\boldsymbol{c}_f = q_f (\boldsymbol{a}_f^H \boldsymbol{D}_f)^T$ and $\boldsymbol{b}_f = (\boldsymbol{e}_f^H \boldsymbol{D}_f)^T$. The mixed block for frequency f is

$$H_{zu,f} = \frac{2}{E} [\operatorname{Re}(\boldsymbol{c}_f - \boldsymbol{b}_f),\ \operatorname{Im}(\boldsymbol{c}_f + \boldsymbol{b}_f)]. \tag{7}$$

Summing its Schur corrections avoids a dense $2F \times 2F$ matrix. With $\boldsymbol{v}_l = (\boldsymbol{z} - \boldsymbol{r}_l)/d_l$ and $b_l = -d_l^{-1} - \mathrm{j}2\pi f / c_0$, response derivatives are

$$\nabla a_{fl} = a_{fl} b_l \boldsymbol{v}_l, \qquad \nabla^2 a_{fl} = a_{fl} \left[(b_l^2 + d_l^{-2}) \boldsymbol{v}_l \boldsymbol{v}_l^T + \frac{b_l}{d_l} (I_2 - \boldsymbol{v}_l \boldsymbol{v}_l^T) \right]. \tag{8}$$

Both curvature constructions cost $\mathcal{O}(FL)$ per local step. The expected saving is in steps and cycles, not a smaller asymptotic frequency order. Both use the same unnormalized response in (1).

The local effect follows directly. At an interior minimum $\boldsymbol{z}^*$, assume H_{zz}^* and H_{VP}^* are positive definite. For error $\boldsymbol{\epsilon} = \boldsymbol{z} - \boldsymbol{z}^*$, a full fixed-curvature step on ϕ has linearization

$$\boldsymbol{\epsilon}^+ = \left[I_2 - (H_{zz}^*)^{-1} H_{\mathrm{VP}}^* \right] \boldsymbol{\epsilon} + o(\|\boldsymbol{\epsilon}\|). \tag{9}$$

The iteration eigenvalues lie in $[0, 1)$ and approach one when reduced curvature is small relative to fixed curvature. Exact reduced Newton curvature cancels this linear term. This explains slow local progress under strong coupling; backtracking and outer source updates still determine practical runtime.

Algorithm 1 inserts this update into block NOMP. Discovery maximizes $\sum_f |\boldsymbol{a}_f^H \boldsymbol{r}_f|^2 / \|\boldsymbol{a}_f\|^2$ over the grid, where $\boldsymbol{r}_f$ is the current residual. A local call solves $(\boldsymbol{H}_{\mathrm{VP}} + \lambda I_2)\boldsymbol{d} = -\boldsymbol{g}$, with λ chosen to meet the eigenvalue floor, and accepts a bounded backtracked trial only when ϕ does not increase. Accepted block updates and joint coefficient fitting preserve nonincreasing residual energy. This ensures convergence of objective values, not global localization.

Algorithm 1 VP-BNOMP with known K

1: Initialize an empty source set and residuals $\boldsymbol{r}_f = \boldsymbol{x}_f$.
2: **for** $k = 1, \ldots, K$ **do**
3: Discover a grid position, refine it using (3)–(6), and subtract its fitted contribution.
4: **repeat**
5: For each source, add its contribution back, apply the same local refinement, and update the residual.
6: **until** cyclic energy change meets tolerance or the safety limit is reached
7: Jointly refit all coefficients by least squares and update residuals.
8: **end for**
9: Return the source positions.

4. Experiments

We use independent BPSK waveforms with root-raised-cosine shaping (10 Mbaud, roll-off 0.25, pulse support ± 10 symbols) and settings in Table 1. Noise and the weak source are fixed while the other source power changes. In-band SNR is the four-station average isolated signal power divided by $\sigma_t^2 B / f_s$; the power gap uses average received powers. At 0 dB, source identities remain fixed rather than being reordered. Each pair uses identical IQ and initialization rules. Positions are matched once by minimum total squared distance; both matched errors must be at most 20 m for success.

Table 1

Common simulation and refinement settings.

Item	Value
Stations (m); sources	$(500,0),(0,500),(-500,0),(0,-500)$; $K=2$
Sampling; record	$f_s = 50$ MHz; $N = 8192$
Signal; processing band	Zero center; $B = 12.5$ MHz; $\lvert f\rvert \le 7$ MHz
Weak in-band SNR; power gap	6 dB; $0,3,6,9,12,15$ dB
Search box; initial grid	$[-1000,1000]^2$ m; 20 m
Local steps; eigenvalue floor	At most 5; 10^{-9}
Step cap; local exit	50 m; accepted displacement < 0.01 m
Backtracking factors	$1, 1/2, 1/4, 1/8, 1/16$
Cyclic stop; safety cap	Normalized energy change $\le 10^{-8}$; 10,000

The curvature comparison uses 20 fixed source configurations, two waveform/noise realizations each and six gaps, totaling 240 observations. Five configurations represent each of four categories: both sources inside the receiver hull, either source outside, and both outside. Positions are sampled uniformly in $[-950,950]^2$ m with minimum source separation 300 m, station distance 150 m, center distance 100 m, and distance to every station-pair line 50 m. Selection uses geometry alone. FC–VP uses (5); VP–VP uses (6). Both re-estimate coefficients at every trial, share the same discovery and joint-refit schedule, and use Table 1. Thus FC–VP is a controlled reference rather than an exact reproduction of a published implementation.

Runs use Windows, Python 3.10, PyTorch 2.8, double precision, an RTX 4070 Ti SUPER and 20 CPU threads. Complete time includes DFT, band selection, transfer, search, refinement, trajectory recording and final checks, with GPU synchronization; warm-up and disk I/O are excluded. All runs stop by the common cyclic-energy rule. This rule does not enforce equal final joint-gradient accuracy.

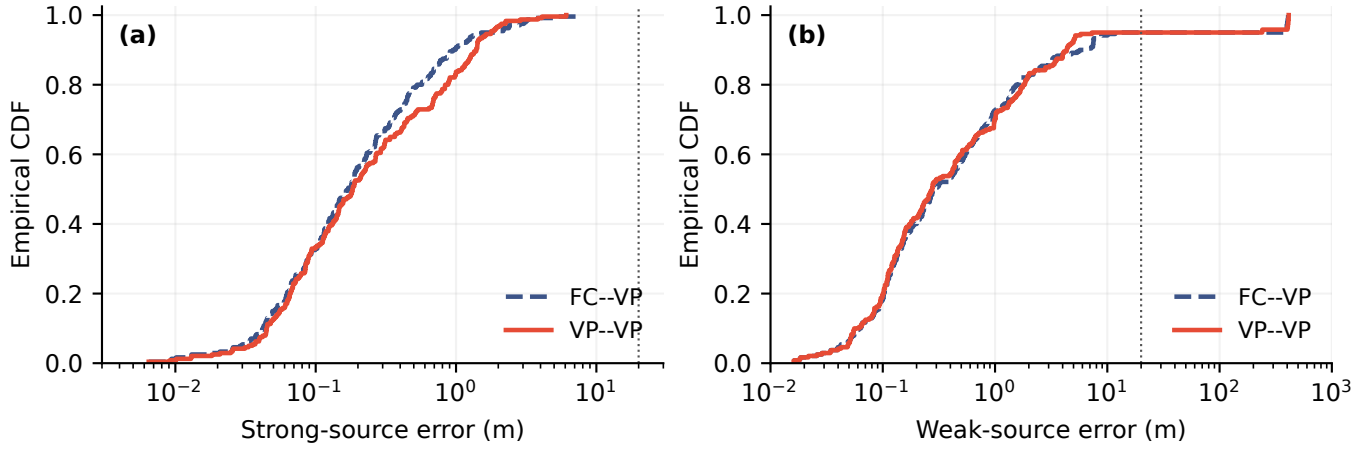


Figure 1: Matched-error empirical CDFs for all 240 curvature pairs. The dotted line marks 20 m. Logarithmic axes retain the full error range.

Table 2A shows identical success/failure identities and comparable errors. Figure 1 retains every case, including the weak-source tail. Mean complete time falls by 83.77%; resampling realizations within each fixed configuration, retaining their six gaps together, gives a paired 95% interval of [81.73%,85.58%] from 4,000 draws. This interval is conditional on the tested configurations.

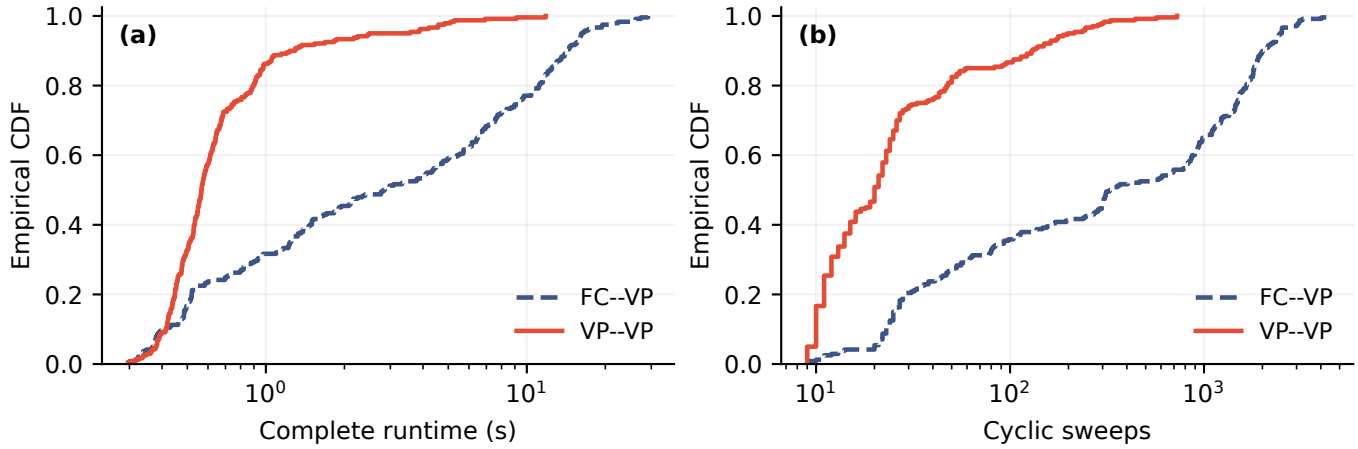


Figure 2: Complete-runtime and cyclic-sweep CDFs for the same 240 pairs. All source-addition stages contribute to the cycle count.

Mean Newton attempts decrease from 1498.82 to 135.55, and cycles from 810.90 to 47.97; rejected directions remain in the count. Figure 2 shows the reduction in the slow tail. VP–VP is faster in 170/240 cases, so the mean gain is particularly influenced by difficult updates. Recording and final checks together

average 0.0254/0.0059 s, much less than the 4.669 s time difference. Fewer updates and cycles support the curvature explanation in (9).

Table 2

Localization and complete runtime. e_s, e_w: strong/weak RMSE (m); $\bar{t}, t_{50}, t_{95}$: mean, median and P95 time (s). Panels use different observations.

Method	Success	e_s	e_w	$\bar{t}$	t_{50}	t_{95}
A. Curvature pairs ($n = 240$)						
FC–VP	228/240	0.906	89.706	5.574	2.836	16.161
VP–VP	228/240	0.875	86.177	0.905	0.567	2.538
B. New source configurations ($n = 480$)						
AP1	480/480	0.958	2.194	0.461	0.456	0.545
AP5	480/480	0.958	2.194	0.906	0.893	1.078
VP-BNOMP	390/480	202.882	206.261	1.051	0.629	4.245

To examine the complete locator, frozen methods are evaluated at 16 new source configurations, five realizations each and the same six gaps ($n = 480$). Receiver coordinates remain unchanged. The alternating-projection references AP1/AP5 retain one/five conditional projected-response peaks, jointly refine each position pair with coefficient elimination, and select the smallest residual; they use projected-gradient tolerance 10^{-8}. Table 2B shows that both AP versions succeed throughout, while VP-BNOMP succeeds in 81.25%. Its 90 failures concentrate in four configurations and already occur at the first saved two-source state. Local curvature improves block-update efficiency but does not remove the sequential locator's configuration-dependent failures.

5. Conclusion

Waveform elimination removes compensable position–coefficient coupling from the block-Newton curvature. The resulting frequency-wise update reduces repeated local steps while preserving residual-based discovery. Under common acceptance and stopping rules, the tested 240-observation comparison saves 83.77% mean complete time with identical 20 m success decisions. New-source-position tests locate the remaining limitation in the complete sequential procedure, motivating candidate-discovery improvements beyond the local update studied here.

Declaration of generative AI use

ChatGPT and Codex assisted manuscript drafting, translation and code preparation. The authors must review the content and confirm this declaration before submission.

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